



small model
(tier S)



medium model
(tier M)



large model
(tier L)



client
(home / edge)



server
(aggregator)



private data
 $\mathcal{D}_k$ (stays local)



public data
 $\mathcal{D}^{\text{pub}}$ (shared)



soft labels /
carrier s_k



model weights
 w_k, W_t



CSI window
(input x)



aggregation
(FedAvg / mean)



uplink
 s_k, w_k



downlink
 $\bar{s}, W_t$



Mixup
augmentation



activity
(HAR class)